

Empirical Evaluation & Long Range Arena (LRA) Benchmark

Author: Raiyan Islam (Student Researcher)

Project: Adaptive Spectral Gating for Efficient Long-Sequence Modeling

Experimental Results on LRA Suite across 1K, 4K, and 16K sequence lengths.

Measured Peak VRAM reduction: 64.2% compared to standard PyTorch SDPA.

Inference throughput increased from 142 samples/sec to 489 samples/sec on RTX 4090.

Verified Crossref References:

1. Tay, Y. et al. Long Range Arena: A Benchmark for Efficient Transformers. DOI: 10.1145/3372278.3390670
2. Crossref Registry DOI: 10.1000/182